Paul Parisot

Software Engineer - C++ & Low-Latency Systems

Github: paulogarithm — paul.parisot@epitech.eu - +33 6 42 36 01 57

About

Systems software engineer deeply focused on modern C++ performance optimization, low-level Linux runtimes, and hardware-software boundaries.
Specialized in zero-allocation memory design, concurrent thread synchronization, and asynchronous networking architectures.
Pragmatic debugger experienced in diagnosing subtle performance bottlenecks, memory corruptions, and race conditions in production environments.

Skills

Core Systems: C++ (Modern C++17/20), C, Go, Python, x86/ARM Assembly stubs
Low-Level Linux: Memory Management (`heap, stack, shm`), POSIX Threads, Atomics, Driver Communication Boundaries
Performance & Networking: Asynchronous I/O (`Boost.Asio`), TCP/IP Paradigms, Cache-Line Linearity (ECS Layout)
Engineering Toolchain: Linux, CMake, GCC, GDB, Valgrind (Memcheck/Helgrind), Git, Jenkins CI/CD pipelines

Professional Experience

Faurecia / Forvia - Software Engineer (Systems & Tooling)

- Developed and optimized low-level C++ software modules tailored for hardware driver communication interfaces.
- Engineered internal testing & visual tracking applications using C# (Unity engine) under strict system resource limitations.
- Analyzed physical architecture constraints to maintain deterministic software behavior and memory guarantees.
- Collaborated closely within an industrial Agile/Scrum framework, tracking requirements and features via **Jira**.

Gno.land - Core Software Engineer (Distributed Runtime Engines)

- Refactored core multi-threaded execution sub-systems to guarantee operational determinism and maximum execution speed.
- Isolated, reproduced, and patched **6 critical bugs** causing runtime crashes, delivering 10k+ LOC to production.
- Authored automated regression and integration testing suites embedded into continuous deployment (CI/CD) workflows.

Epitech / IONIS - Systems Programming Instructor & Code Auditor

- Taught advanced C/C++ memory architectures, pointer arithmetic, and native Linux system programming to **400+ students**.
- Audited large-scale repositories to enforce strict production standards: error-handling paths, cache awareness, and memory safety.

Low-Latency & Systems Technical Projects

rodox – Low-Latency Linux Systems Engine

Built a modular runtime infrastructure in C++ using a data-oriented Entity Component System (ECS) to avoid cache misses.
Completely bypassed OS-level heap allocation latencies (`malloc`/`new`) by designing thread-safe **custom static pool allocators**.
Implemented high-throughput, non-blocking asynchronous networking components utilizing `Boost.Asio` over TCP. Hardened execution paths against hardware-software latency spikes using profiling tools including **CMake**, **GDB**, and **Valgrind**.

Skills: Modern C++, SIMD, Boost.Asio, Memory Management, Linux System APIs, Profiling (Valgrind), Concurrency

cclass – Runtime Memory Mutation Engine

Engineered a raw C engine capable of dynamically rewriting compiled assembly instructions inside secure execution memory regions.
Leveraged system-level `mmap` calls with explicit page protections to remove virtual dispatch runtime overheads.
Enforced strict memory alignment checks to prevent hardware execution faults across target host platforms.

Skills: C, Linux Systems, x86 Assembly, mmap, Memory Safety, Low-level Debugging

Sledge – Containerized Orchestration & Core Backend

Architected and implemented a high-performance core backend infrastructure in Go optimized for high-volume execution streams.
Utilized the Docker API at runtime to dynamically spin up, isolate, and orchestrate concurrent environment containers.
Wrote automated validation and monitoring tooling using **Python** scripts to verify continuous data flow and system state health.

Skills: Go, Docker API, Python Scripting, Data Streaming, System Automation

Education

Epitech, Paris 2022 – 2027
Master's Degree in Information Technology (Master's level diploma)
Level-7 Certification - Expert in Software Engineering
Specialization: Systems Architecture

McGill University, Montréal 2025 – 2026
Certificate in Management
Technical Leadership & Strategy